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Semester: III	Year : AY 2026-27
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Subject: Programming with Java

ASSIGNMENT No : 15

TITLE : Exception Handling using Finally, Throw, and Throws

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Theory

The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

Programme

```
import java.util.Scanner;

public class Login {

    static void checkPassword(String password) {
        if (!password.equals("12345")) {
            throw new IllegalArgumentException("Invalid password!");
        }

        System.out.println("Login successful.");
    }

    public static void main(String[] args) {
```

```

Scanner sc = new Scanner(System.in);

try {
    System.out.print("Enter password: ");
    String password = sc.nextLine();

    checkPassword(password);
}
catch (IllegalArgumentException e) {
    System.out.println("Error: " + e.getMessage());
}
finally {
    System.out.println("Login process completed.");
}

sc.close();
}
}

```

```

import java.util.Scanner;

public class ATM_PIN {

    static void verifyPIN(int pin) throws Exception {
        if (pin != 1234) {
            throw new Exception("Invalid PIN!");
        }

        System.out.println("PIN verified successfully.");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

```

```

try {
    System.out.print("Enter ATM PIN: ");
    int pin = sc.nextInt();

    verifyPIN(pin);
}
catch (Exception e) {
    System.out.println("Error: " + e.getMessage());
}
finally {
    System.out.println("PIN verification process completed.");
}

sc.close();
}
}

```

Output

```

● rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 15 % cd "/Users/rakesh"
● rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 15 % cd "/Users/rakesh"
Enter password: rakesh
Error: Invalid password!
Login process completed.
○ rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 15 % █

```

```

Problems  Output  Debug Console  Terminal  Ports
● rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 15 % cd "/Users/rakesh"
● rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 15 % cd "/Users/rakesh"
Enter ATM PIN: 1234
PIN verified successfully.
PIN verification process completed.
⊗ rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 15 % 2356
zsh: command not found: 2356
🔗 rakeshkumarchaudhary@RAKESHs-MacBook-Air Experiment 15 % █

```

Conclusion

The program successfully demonstrates the use of `throw`, `throws`, and `finally` for effective exception handling and proper execution of cleanup or completion statements.